BioMed-Scribe: Fine-Tuning GPT-2 for Medical Specialty Classification from Clinical Conversations

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A Comprehensive Project Report

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1 Abstract

BioMed-Scribe is a four-class medical specialty classifier built by fine-tuning GPT-2 Medium on doctor-patient conversations. The task is to assign each conversation to Cardiology, Neurology, Orthopedics, or General Medicine. Because the two public source datasets do not provide specialty labels, a keyword-based labeling pipeline was used to create a balanced working corpus. After filtering, labeling, sampling, and balancing, the final dataset contained 13,492 conversations, including 10,793 training examples. The final model unfroze the last two GPT-2 transformer blocks rather than only the final block, increasing the number of trainable parameters from 12.6 million to 25.2 million, or 7.1% of the 354.8 million total parameters. The model was trained for 14 epochs on an NVIDIA A100 GPU using a two-phase learning-rate schedule. The best checkpoint achieved 79.1% test accuracy and a macro F1 score of 0.799. Neurology had the highest recall at 93.6%, while General Medicine remained the most difficult class because of its broad and overlapping symptom vocabulary. Compared with earlier single-block versions, the two-block model improved overall accuracy, macro F1, and every class-specific F1 score. A t-SNE analysis of final hidden-state representations showed partial specialty-specific structure, and a Gradio demonstration correctly classified 9 of 10 unseen examples. Overall, the project shows that limited fine-tuning of a general-purpose language model can produce useful specialty classification performance, while also showing that label quality and class definition are major constraints on further improvement.

2 Introduction

Clinical conversations contain information that can help determine the type of specialist most appropriate for a patient. In telehealth and other high-volume settings, automated specialty classification could support routing and triage by converting free-text conversations into a small set of clinically relevant categories. This project studies whether a general-purpose language model can learn that classification task without full-model fine-tuning.

GPT-2 is a decoder-only Transformer model that learns contextual text representations through self-attention. Rather than retraining all 354.8 million parameters of GPT-2 Medium, BioMed-Scribe uses parameter-efficient fine-tuning: most of the pretrained network is frozen and only the final transformer blocks and classification head are updated. The main experiment evaluates whether unfreezing the final two transformer blocks provides a measurable improvement over the earlier single-block approach.

The project addresses four questions:

1. Can GPT-2 Medium distinguish among Cardiology, Neurology, Orthopedics, and General Medicine from clinical conversation text?
2. Does two-block fine-tuning improve performance relative to one-block fine-tuning?
3. Which specialties are easiest or hardest to classify, and where do errors occur?
4. Do the learned hidden-state representations show specialty-specific structure?

The final system combines a data-labeling pipeline, a two-block GPT-2 classifier, error and embedding analyses, and an interactive Gradio interface.

3 Background

3.1 Transformer and GPT-2 Representation

The Transformer architecture replaces recurrent sequence processing with attention. Scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{1}$$

where Q, K, and V are query, key, and value matrices. Multi-head attention applies this operation across several learned projection spaces, allowing the network to represent different relationships within the same sequence.

GPT-2 Medium contains 24 decoder blocks, a hidden dimension of 1024, 16 attention heads per block, and approximately 354.8 million parameters. For classification, the language-modeling output is replaced by a four-class linear head. The last non-padding token is used as the sequence representation because, under causal masking, that token has access to all preceding tokens in the conversation.

3.2 Why Partial Fine-Tuning?

Fine-tuning only upper transformer blocks reduces computation while preserving lower-layer language representations. Upper layers are closer to the task output and can adapt more directly to specialty-specific wording. The final version therefore unfreezes blocks `h[22]` and `h[23]`, together with the final layer normalization and classification head, while blocks 0–21 remain frozen.

4 Data and Label Construction

4.1 Data Sources

Two public HuggingFace datasets were used. The primary source, the AGBonnet augmented clinical notes dataset, contains augmented doctor-patient conversations and structured clinical summaries. After filtering for minimum length and dialogue structure, 22,068 usable conversations remained. The second source, the ChatDoctor HealthCareMagic dataset, contains patient-doctor question-answer pairs; these were converted to a conversation format and sampled to supplement the specialty classes.

Neither source provides the four specialty labels required for this project. Labels were therefore assigned using a rule-based keyword system. Each conversation was scored according to specialty-specific vocabulary, and the specialty with the largest number of matches was assigned. Ties or conversations without a clear specialty signal defaulted to General Medicine.

Table 1: Examples of keywords used by the specialty-labeling rules.

Specialty	Selected terms
Cardiology	chest pain, palpitations, angina, ECG, hypertension, coronary, arrhythmia, myocardial, atrial fibrillation, stent
Neurology	headache, migraine, seizure, stroke, numbness, dizziness, tremor, neuropathy, paralysis, speech difficulty
Orthopedics	knee pain, hip pain, fracture, joint pain, spine, back pain, ligament, tendon, arthritis, sprain
General Medicine	fever, cough, fatigue, nausea, vomiting, diarrhea, rash, infection, diabetes, thyroid, abdominal pain

4.2 Final Analytic Dataset

After balancing and supplementation, the combined corpus contained 13,492 examples. A stratified split produced 10,793 training examples, 1,349 validation examples, and 1,350 held-out test examples. General Medicine was the largest class and Neurology the smallest. Because the original report gives 267 Neurology examples in the split table but 266 in the final classification output and confusion matrix, the evaluation results in this paper follow the final model output (266 Neurology test cases).

Table 2: Final dataset size used for model development.

Split	Examples	Purpose
Training	10,793	Parameter updates
Validation	1,349	Model selection and checkpointing
Test	1,350	Final held-out evaluation
Total	13,492	Final labeled corpus

The labeling procedure is also the main methodological limitation of the project. Since labels are generated from the same kinds of symptom words that the classifier later sees, test accuracy measures agreement with the constructed labeling system rather than agreement with physician-assigned specialty labels.

5 Model and Training Procedure

5.1 Two-Block GPT-2 Classifier

Let $H \in \mathbb{R}^{B\times T\times 1024}$ denote the final hidden states for a batch and $M \in \{0,1\}^{B\times T}$ the attention mask. If

$$\ell_b = \sum_t M_{b,t} - 1$$

is the final non-padding token for sequence b, the class probabilities are

$$\hat{y}_b = \operatorname{softmax}\left(W \operatorname{Dropout}(H_{b,\ell_b}) + c\right), \tag{2}$$

where $W \in \mathbb{R}^{4\times 1024}$.

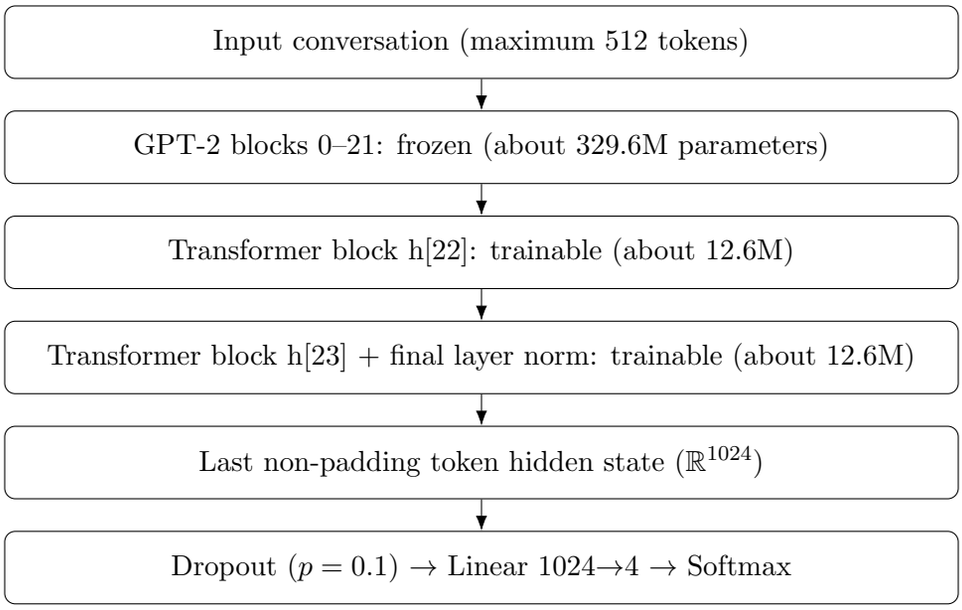


Figure 1: BioMed-Scribe two-block fine-tuning architecture. Only the final two GPT-2 blocks, final normalization, and classification head are updated.

Table 3: Parameter allocation in the final model.

Component	**Parameters**	**Trainable?**
GPT-2 blocks 0–21	329.6M	No
Block h[22]	12.6M	Yes
Block h[23] + final layer norm	12.6M	Yes
Classification head	4.1K	Yes
Total	354.8M	25.2M (7.1%)

5.2 Optimization

Training was performed in Google Colab on a single NVIDIA A100 SXM4-40GB GPU. Inputs were tokenized with GPT-2 byte-pair encoding and padded or truncated to 512 tokens. Batch size was 16 with gradient accumulation over two steps, giving an effective batch size of 32. A `WeightedRandomSampler` was used to reduce the effect of class imbalance.

AdamW optimization used weight decay 0.01 and gradient clipping at norm 1.0. Training was divided into two phases: epochs 1–5 used a learning rate of 3×10^{-5} with linear warmup, and epochs 6–14 used 1×10^{-5}. Float32 training was retained because the attempted mixed-precision configuration produced unstable NaN losses. The checkpoint with the best validation accuracy was saved automatically.

Table 4: Selected training history.

Epoch	Train Loss	Val Loss	Train Acc.	Val Acc.	Macro F1
1	2.004	0.947	37.9%	63.0%	0.624
2	0.757	0.683	70.7%	73.5%	0.742
3	0.580	0.603	77.7%	77.5%	0.784
5	0.501	0.575	81.1%	77.8%	0.785
7	0.463	0.553	82.5%	79.3%	0.800
9	0.422	0.532	83.9%	80.3%	0.810
12	0.418	0.523	84.0%	**80.7%**	**0.814**
14	0.412	0.520	84.8%	80.5%	0.812

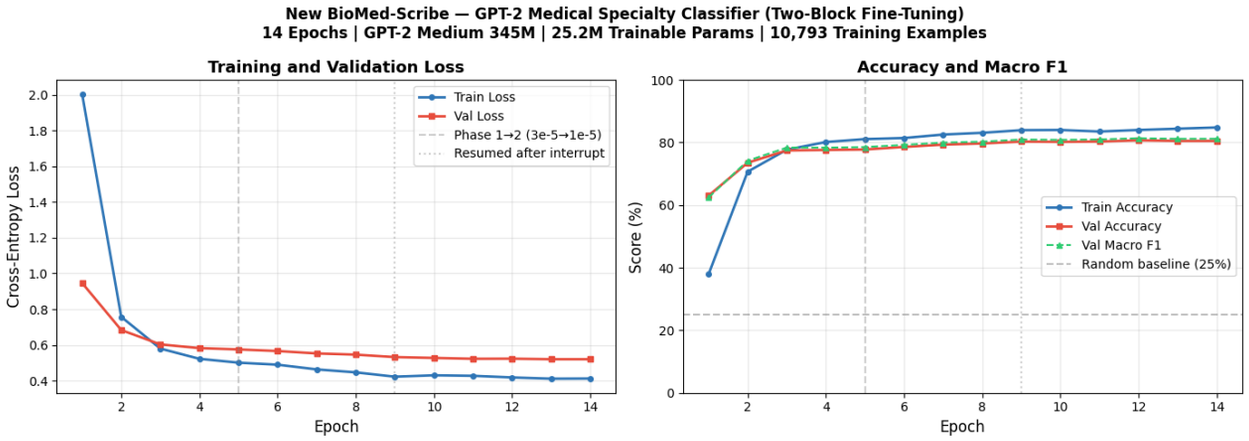


Figure 2: Training and validation loss, accuracy, and macro F1 across 14 epochs. Validation performance levels off near epochs 12–14 while training and validation curves remain close.

6 Results

6.1 Held-Out Test Performance

The best checkpoint achieved 79.1% accuracy on 1,350 held-out examples with a macro F1 score of 0.799. Neurology was the strongest class, with 93.6% recall and F1 of 0.894. General Medicine was the weakest class, with 65.0% recall and F1 of 0.697.

Table 5: Classification performance on the held-out test set.

Specialty	**Precision**	**Recall**	**F1**	**Support**
Cardiology	0.815	0.857	0.835	314
Neurology	0.856	0.936	0.894	266
Orthopedics	0.757	0.784	0.770	370
General Medicine	0.751	0.650	0.697	400
Macro average	0.795	0.807	0.799	1,350
Weighted average	0.788	0.791	0.788	1,350

6.2 Improvement Across Model Versions

The project developed three successive versions. Version 1 used one trainable block for three epochs. Version 2 retained one trainable block but increased training to 15 epochs and the token limit to 512. Version 3, the final model, unfroze two blocks. Performance improved at every stage.

Table 6: Comparison of the three BioMed-Scribe model versions.

Metric	v1: 1 block	v2: 1 block	v3: 2 blocks
Test accuracy	74.5%	77.3%	**79.1%**
Macro F1	0.743	0.782	**0.799**
Cardiology F1	0.797	0.822	**0.835**
Neurology F1	0.819	0.877	**0.894**
Orthopedics F1	0.693	0.758	**0.770**
General Medicine F1	0.663	0.673	**0.697**
Trainable parameters	12.6M	12.6M	25.2M
Epochs	3	15	14
Maximum tokens	256	512	512

The final two-block configuration improved test accuracy by 1.8 percentage points over v2 and by 4.6 points over v1. Because v2 and v3 differ in more than one training detail, the comparison is best interpreted as an empirical project progression rather than a controlled causal estimate of the effect of unfreezing one additional block.

6.3 Error Analysis

The confusion matrix shows that most remaining errors involve General Medicine. Orthopedics is predicted as General Medicine 53 times, while General Medicine is predicted as Orthopedics 72 times. Cardiology also leaks into General Medicine in 27 cases. By contrast, Neurology has only 17 misclassified cases among the 266 examples in the final evaluation output.

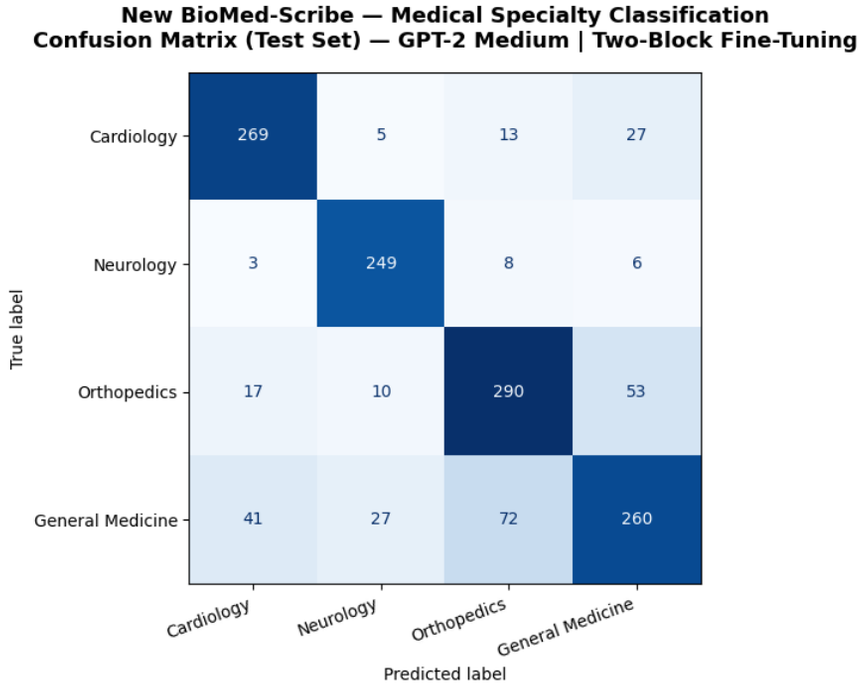


Figure 3: Confusion matrix for the held-out test set. The largest bidirectional confusion occurs between Orthopedics and General Medicine.

Table 7: Confusion matrix counts. Rows are true classes and columns are predicted classes.

	Cardio	Neuro	Ortho	Gen. Med.
Cardiology	269	5	13	27
Neurology	3	249	8	6
Orthopedics	17	10	290	53
General Medicine	41	27	72	260

These errors are consistent with the labeling design. Neurological terms such as *seizure*, *stroke*, and *migraine* are relatively distinctive, while General Medicine is a broad default category that shares symptoms such as pain, fatigue, dizziness, and nonspecific discomfort with the other specialties. Thus, some classification ambiguity is built into the task definition itself.

7 Embedding Analysis

To examine the model representation directly, the 1024-dimensional final hidden state was extracted for 500 test examples and projected to two dimensions using t-SNE with perplexity 30 and PCA initialization. These are the same hidden-state vectors used by the classifier head.

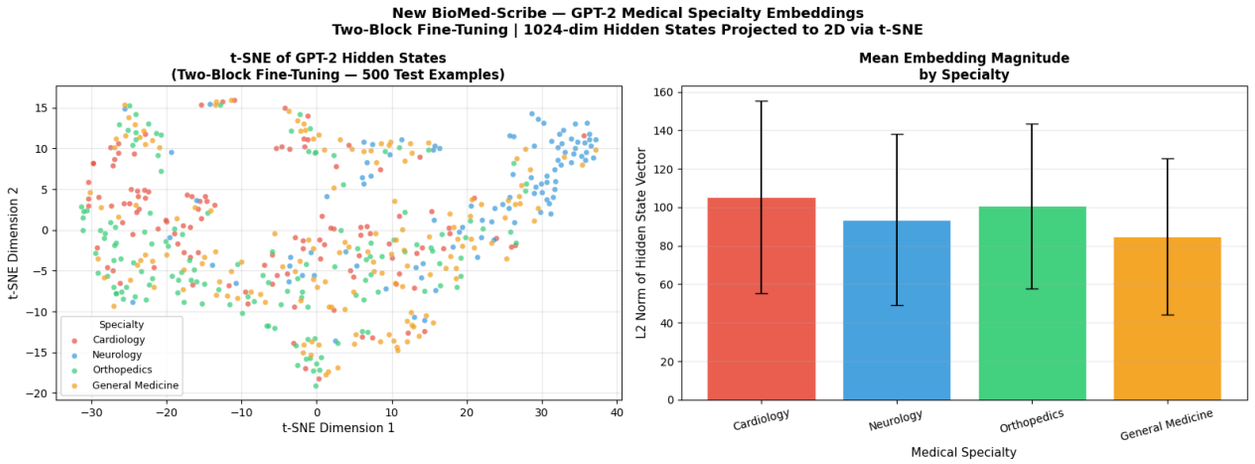


Figure 4: t-SNE projection of final hidden states (left) and mean hidden-state L2 magnitude by specialty (right). The projection shows partial specialty-specific structure but substantial overlap.

Neurology and Orthopedics show partial separation, while Cardiology and General Medicine overlap more strongly. This pattern agrees qualitatively with the confusion matrix. Mean L2 norms are 105.4 for Cardiology, 93.5 for Neurology, 100.6 for Orthopedics, and 84.8 for General Medicine. The range of mean norms is smaller than in the earlier one-block model, but the two-dimensional t-SNE display should be interpreted descriptively: t-SNE is a nonlinear projection and does not by itself prove that the underlying 1024-dimensional classes are cleanly separated.

8 Deployment

The final checkpoint was deployed in an interactive Gradio application. A user enters a doctor-patient conversation, the text is tokenized to a maximum of 512 tokens, and the interface reports the predicted specialty together with confidence scores for all four classes. A low-confidence warning is shown when the top probability is below 50%, encouraging manual review of ambiguous cases.

A demonstration using ten unseen examples produced nine correct specialty predictions. The single incorrect example involved palpitations together with dizziness and was predicted as Neurology rather than Cardiology with only 39% confidence. The low confidence is useful operationally because it indicates that the case lies near an ambiguous decision boundary.

Table 8: Summary of the ten-example live demonstration.

Outcome	Count	Rate
Correct specialty	9	90%
Incorrect specialty	1	10%
Low-confidence ambiguous error	1	10%

The demonstration is useful as a software proof of concept, but ten hand-selected examples are too few to provide a statistically reliable estimate of real-world accuracy. The held-out test set remains the more meaningful quantitative evaluation.

9 Discussion

The central result is that a relatively small trainable portion of GPT-2 Medium can learn a four-specialty classification task from clinical conversation text. Updating only 7.1% of model parameters produced 79.1% test accuracy and macro F1 of 0.799. The progression from v1 to v3 also shows that longer context, longer training, and additional trainable capacity were associated with steady improvements in all reported class-level F1 scores.

The results also show that task formulation matters as much as model architecture. Neurology performs best because many neurological presentations contain highly discriminative vocabulary. General Medicine performs worst because it functions as a broad residual class and shares substantial vocabulary with the other categories. The most important remaining errors are therefore not random: they cluster where the label definitions themselves overlap.

The two-block model is more computationally expensive than the one-block model but still far cheaper than full fine-tuning. This makes it a reasonable compromise for a course-scale project using a single GPU. The model also provides a useful example of how transfer learning can be adapted to a practical classification problem without training a language model from scratch.

9.1 Limitations

The strongest limitation is label quality. Specialty labels were generated by a keyword rule rather than assigned by physicians. This creates at least three concerns. First, some conversations may receive incorrect labels. Second, the labeling process can amplify the same vocabulary cues that the model later learns, making evaluation partly circular. Third, the default General Medicine label produces a heterogeneous class whose boundaries are less clinically precise.

The task also covers only four specialties, while real clinical routing involves many more categories. The source datasets differ from actual hospital transcripts, so external performance is unknown. In addition, GPT-2 was not pre-trained specifically on clinical text. Finally, the three model versions were developed sequentially rather than through a tightly controlled ablation study, so differences between versions should not be attributed exclusively to the number of unfrozen blocks.

9.2 Future Work

The most valuable next step is expert annotation. Even a smaller corpus of physician-labeled conversations would provide a more defensible target and a more meaningful external evaluation. A second priority is to expand the taxonomy to specialties such as Psychiatry, Dermatology, and Gastroenterology. Controlled ablation experiments could then compare one-, two-, and three-block fine-tuning under identical data splits, token lengths, learning-rate schedules, and epoch budgets. Evaluation on authentic electronic health record or triage transcripts would be necessary before any clinical use could be considered.

10 Conclusion

BioMed-Scribe demonstrates that GPT-2 Medium can be adapted to medical specialty classification using partial fine-tuning rather than full-model training. The final two-block model

updates 25.2 million parameters and achieves 79.1% held-out accuracy with a macro F1 score of 0.799. Neurology is classified most reliably, while General Medicine remains challenging because it is broad and overlaps with other specialties. Embedding and confusion-matrix analyses tell a consistent story: the model learns meaningful specialty structure, but the decision boundaries are blurred where symptom language overlaps.

The project therefore supports two conclusions. First, modest fine-tuning capacity is sufficient to extract useful task-specific performance from a general pretrained language model. Second, further progress is likely to depend less on adding parameters than on improving label quality, class definitions, and evaluation data. In that sense, BioMed-Scribe is both a successful classification experiment and a demonstration of a broader principle in applied machine learning: strong models cannot fully compensate for an ambiguous target.

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